

KYLE J. UTLEY

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Mechanical Engineering Student | Systems Design & Validation Engineering

EDUCATION

The University of Alabama
Tuscaloosa, AL — Bachelor Of Science: Mechanical Engineering

Expected Graduation May 2028

- Cumulative GPA: 3.67 | Member, University of Alabama Honors College | Accelerated STEM-to-MBA Program

RELEVANT EXPERIENCE

EQUIPMENT ENGINEERING CO-OP – LINE 1
Honda Motors | Lincoln, AL

Aug 2024 - Dec 2024 | Jan 2025 - Apr 2025

- Modeled a new structural lift-assist fixture in CATIA, applying DFM principles, then fabricated the frame in-house with a plasma cutter, lathe, and mill, joined via MIG welding; consolidated pneumatic controls, cut unit weight by 7.0 lb, and added rubber-cushioned motion limiters to reduce operator strain.
- Achieved zero unplanned downtime since installation and positive operator feedback, with a more serviceable design that simplified ongoing maintenance and troubleshooting.
- Diagnosed a recurring sensor jam as a mechanical design flaw, iterating actuator flag material (metal, nylon, Delrin) to eliminate wear; redesigned switches have had zero jams since installation, eliminating a manual-handling workaround that risked part damage and operator injury.
- Manufactured in-house replacement parts for worn equipment components, avoiding costly expedited outsourcing to external machine shops and meaningfully reducing part-replacement spend.

SUSPENSION TEAM MEMBER
Crimson Racing – Formula SAE, The University of Alabama | Tuscaloosa, AL

Jan 2024 - Present

- Designed a suspension linkage component (bellcrank) in SolidWorks for a formula-style vehicle platform, validating geometry via FEA against defined structural load cases and machining the final part from 7075-T6 aluminum to minimize mass while preserving structural margin.
- Component passed a full season of structural validation under real operating loads without a single failure, confirming design margin and manufacturing quality.
- Designed a precision-slotted rotational reference component and magnetic sensor mounting fixture to develop a new rotational speed detection method, manufacturing the part via waterjet to hold tight installation tolerances.
- Validated the new sensing method through extensive field testing, cross-referencing measured data against an existing reference system to confirm accuracy, establishing the method for use across multiple vehicle platforms.

ADDITIONAL EXPERIENCE

STUDENT ASSISTANT
Manderson Graduate School of Business, The University of Alabama | Tuscaloosa, AL

July 2026 - Present

- Provide cross-functional support across reception, records management, event logistics, and communications, serving as a primary point of contact for visitors, students, and faculty.

TECHNICAL SKILLS

CAD & Design - SolidWorks CATIA FEA Analysis Fusion 360 - Root Cause Analysis DFMEA DFM Sensor Integration	Shop & Fabrication - CNC Machining 3D Printing Waterjet Plasma Cutter - MIG Welding Lathe Mill Press Grinder Bandsaw
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INVOLVEMENT AND HONORS

- Member, Alabama AI's Pals Mentorship Program - Member, American Society of Mechanical Engineers	- University of Alabama Dean's List Fall 2025 - University of Alabama President's List Spring 2024
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